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Autonomous Robotics & Hardware Acceleration
AMD Zynq UltraScale+ MPSoC (xck26-sfvc784-2LV-c)

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Core System Insight

Offloading neural obstacle avoidance policies to AMD Kria KR260 FPGA Programmable Logic guarantees **deterministic sub-millisecond execution (**$138.3\,\mu$**s)**, **zero-latency hardware fail-closed safety**, and $271\times$ **higher energy efficiency** compared to traditional host CPU control loops.

Heterogeneous System Architecture: PS-PL Partitioning

Key Architectural Insight

High-level robotics perception and networking run on the Linux Processing System (PS), while neural policy evaluation and safety gates execute deterministically at wire-speed on the UltraScale+ Programmable Logic (PL).

Processing System (PS) — ARM Cortex-A53

- **Operating System:** Ubuntu 22.04 LTS (64-bit ARMv8).
- **Direct Bus Access:** Pure Python `/dev/mem` driver via high-speed memory-mapping (Zero C++ dependency).
- **On-Target Management:** Linux FPGA Manager (`xmutil` & `/sys/class/fpga_manager`) for dynamic bitstream loading without reboot.

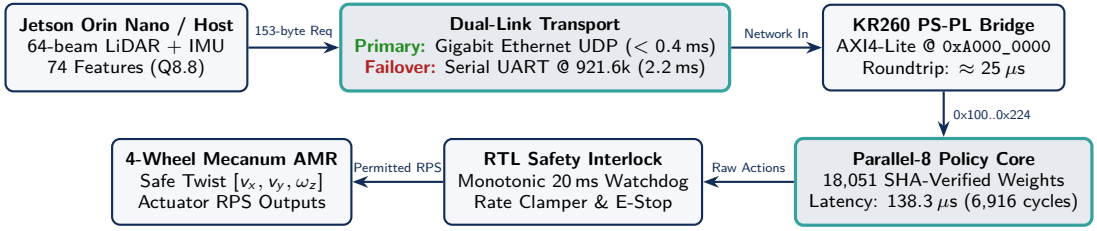
Programmable Logic (PL) — UltraScale+ MPSoC

- **Neural Accelerator:** 8-lane parallel MAC core ($74 \rightarrow 128 \rightarrow 64 \rightarrow 3$ MLP) mapped to 16 DSP48E2 slices and 8 Block RAM banks.
- **Deterministic Core Latency:** Exactly **6,916 clock cycles** (138.32 μs @ 50 MHz) with 40-bit non-overflow accumulators.
- **Hardware Safety Gate:** Autonomous monotonic 20 ms lease watchdog, physical/software E-Stop latch, and rate limiter.

End-to-End Robotics Dataflow & Dual-Link Redundancy

Key Interconnect Insight

A high-speed Gigabit Ethernet UDP link delivers perception data with 10× lower latency than UART, backed by instantaneous, zero-latency hardware UART failover to guarantee unbroken control continuity.



Failover Protection Guarantee

If Ethernet cable disconnects or times out (> 15 ms), the client **seamlessly falls back to UART** without violating the robot's 20 ms deadline.

Multi-Platform Performance & Power Benchmark

Key Benchmark Insight

The AMD Kria KR260 FPGA PL achieves 271× **higher energy efficiency** and **zero OS jitter** compared to host CPUs, completing full neural inference in exactly 138.3 μ**s** at only **0.28 W** dynamic power.

Performance Metric	Laptop Host CPU	KR260 ARM Cortex-A53	KR260 FPGA PL (Ours)
Target Silicon	Intel 12th Gen (x86_64)	Quad ARM Cortex-A53	**AMD UltraScale+ MPSoC**
Compute Clock	3.30 – 4.50 GHz	1.33 GHz	50.0 **MHz (DSP48E2)**
Inference Latency	$\approx 200 - 450\,\mu$s (C/NumPy)	$\approx 450 - 1,200\,\mu$s (NEON)	138.32 μ**s (Exact 6,916 cycles)**
Execution Jitter	$\pm 50 - 500\,\mu$s (OS interrupts)	$\pm 200 - 1,500\,\mu$s (Linux)	0.00 μ**s (Zero Clock Jitter)**
Active Power (Core)	$\approx$ 35.0 W (Package TDP)	$\approx$ 3.78 W (Measured INA260)	0.28 **W (PL Neural Dynamic)**
Energy per Inference	**10, 500** μJ	**3, 024** μJ	**38.7** μJ
Relative Efficiency	1.0× (Baseline)	3.5× more efficient	271× **More Energy Efficient**
Hardware Safety Gate	None (OS Dependent)	None (OS Dependent)	**Fail-Closed Monotonic Watchdog**

Key Takeaways & Field Deployment Readiness

Key Deployment Insight

Project Hephaestus delivers a production-ready, fully turnkey hardware accelerator for autonomous mobile robots with single-command testing, interactive diagnostics, and complete author attribution.

Field-Proven Capabilities

- **100% Bit-Exact Parity:** 16/16 golden test vectors validated against the neural policy oracle.
- **Autonomous Hardware Safety:** Fail-closed 20 ms watchdog cuts motor torque in **0 ns** if communication halts.
- **Redundant Interfaces:** High-speed AXI4-Lite (25 μs) + Gigabit Ethernet ($<$ 0.4 ms) + UART (2.2 ms).

Turnkey Software Integration

- **Interactive Unified Runner:** `./run_kr260.sh` with 10 diagnostic options including live telemetry and latency benchmarks.
- **Automated CI/CD Test:** `make test` and `make test-eth` execute automated headless regressions.
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